



# HOMEWORLD RTX

0.91.3  
PUBLIC BETA 1

Installation, graphics, gameplay options, map authoring, volumetric dust, mission lighting, ray-traced shadows, build notes, and known limitations.

Windows 10/11 x64 - Direct3D 12 / DXR - Non-commercial community project

Requires a legally owned Homeworld 1 Classic installation. Retail game data is not included.

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# Installation and User Guide

## What this build is

Homeworld RTX is a modern Windows x64 port of the original Homeworld engine. It reads the classic game data you already own and supplies a replacement executable, modern renderer, interface assets, and authoring overlays. It does not modify your retail BIG archives.

The beta is intentionally Windows-only. Direct3D 12, DXGI, DXR, the Windows SDK, and NVIDIA Streamline are central to the current renderer. Portability can be revisited after the native renderer reaches campaign-wide parity.

## Installation

### Recommended clean install

1. Download the Windows x64 ZIP from [GitHub Releases](#).
2. Create a writable folder such as `C:\Games\Homeworld RTX`.
3. Extract every file and folder into it.
4. Launch `HomeworldModern.exe`.
5. Choose the `Homeworld.exe` belonging to Homeworld 1 Classic.

The program checks the executable and nearby data folder. The remembered location is stored in:

```
Documents\My Games\Homeworld Modern\installation.ini
```

You can keep Homeworld RTX outside Steam. This protects the mod from Steam verification and keeps exports easy to find.

### Accepted game data

For Homeworld Remastered Collection, select:

```
...\Homeworld\Homeworld1Classic\Homeworld.exe
```

Its data set normally includes:

```
Data\Homeworld.big  
Data\HW_Comp.vce  
Data\HW_Music.wxd
```

For an original 1999 installation, use its `Homeworld.exe` and data directory. `Update.big` is recommended for original data, but should not be added to the Remastered Collection Classic set because its updates are already folded into `Homeworld.big`.

The optional `Movies` directory enables the original pre-mission Bink videos. Retail data is never included in this project's downloads.

### Updating later

Keep personal exports and diagnostics before replacing a beta:

```
RTXExports\  
Documents\My Games\Homeworld Modern\RTXLighting\  
Documents\My Games\Homeworld Modern\Homeworld.cfg
```

Extract a newer release over a copy or into a fresh directory. A fresh folder is easiest to diagnose because obsolete DLLs cannot remain beside the game.

## First run

With no saved profile and no command-line resolution, the game adopts the primary display's active mode and starts borderless. Interface scale is derived from resolution and constrained to the window. Examples:

| Output    | Automatic interface scale |
|-----------|---------------------------|
| 1280x720  | 1.0x                      |
| 1920x1080 | 1.5x                      |
| 2560x1440 | 2.0x                      |
| 3440x1440 | 2.0x centered safe canvas |
| 3840x2160 | 3.0x                      |

The UI, mouse hit regions, fonts, cursor, modal dialogs, and manager docks share one scaling policy. The interface remains native resolution when DLSS is used.

## Video settings

### Display

- **Display Mode:** borderless desktop, exclusive fullscreen, or resizable window.
- **Output Resolution:** applied on the next launch where required.
- **Refresh Rate:** automatic or a supported exclusive-mode refresh.
- **Frame Limit:** presentation cap; zero is uncapped.
- **Interface Scale:** automatic resolution-derived scale with a 50-200% modifier.
- **VSync:** on, adaptive where supported, or off.

Presentation rate does not change simulation rate. AI, physics, scripts, weapons, demos, multiplayer checksums, and save behavior remain tied to the fixed gameplay clock.

### Path tracing

- **Path Tracing:** turns DXR lighting on or off.
- **Light Bounces:** one to three diffuse radiance bounces.
- **Samples:** one to four new stochastic paths per pixel per frame.
- **Generated Normals:** 0-400% LIF-derived micro-surface response.
- **FX Emission:** 0-200% light cast by weapons, engines, beams, explosions,

glow maps, and navigation lights. Zero removes their light, not the visible FX.

Start at one sample and two or three bounces. Increase samples before bounces when you want a steadier image; increase bounces for richer indirect fill. Generated normals at 100% are the tuned baseline.

The temporal accumulator rejects mismatched depth, invalid previous transforms, camera cuts, newly revealed geometry, and LOD/visibility discontinuities. Fast camera movement deliberately retains less history than a still view.

### Image quality

- **Off:** no post anti-aliasing.
- **FXAA:** cross-vendor, inexpensive spatial AA.
- **DLAA:** NVIDIA native-resolution reconstruction.
- **DLSS Quality / Balanced / Performance / Ultra Performance:** NVIDIA

reconstruction modes for the path-light signal.

- **Brightness:** final output transfer.
- **Effect Density and Effect Budget:** control background/trail/impact complexity.
- **Frame Generation:** shown as unavailable; real Streamline DLSS-G and Reflex

are not integrated in this beta.

DLSS does not reduce UI resolution. If an NVIDIA mode is unsupported, use FXAA or Off. DLAA and DLSS require the signed NVIDIA runtime DLLs included with the Windows release and a compatible driver/GPU.

## Post effects

- **Chromatic Aberration:** radial RGB separation; default off.
- **Motion Blur:** uses DXR motion vectors; default off.
- **Luminance Film Grain:** response-controlled grain that protects deep blacks

and highlights; default off.

- **God Rays:** searches the mission background for the authored bright source; ships and planet discs cannot become false suns, and world geometry occludes the radial march.
- **Bloom:** soft scene glow, 0-400%, with 100% as the visual baseline.
- **Color Banding Filter:** stable scene-only output dithering; UI is excluded.

## Lighting model

The ray pass consumes the scene rather than replacing it. It uses:

- Exact visible GEO triangles and object transforms.
- Per-triangle material identity and UVs.
- LIF color, alpha, team masks, and emissive/glow texels.
- Alpha-tested visibility for cutout materials.
- Mission HDR sky radiance and detected bright-source metadata.
- Retail HSF directional, point, spot, and ambient lights.
- Active MEX navigation lights in their transformed ship positions.
- Dynamic engines, layered trails, muzzles, projectiles, harvesting beams,

hyperspace effects, and explosions.

Static geometry is cached in BLAS records. Visible instances are submitted to a triple-buffered TLAS, which is refitted when topology remains stable. Upload buffers and shader tables are reused across frames.

Classic assets do not carry modern metalness/roughness channels. The current model derives conservative material behavior and exposes global reflectivity, roughness, and generated-normal controls. Specular ray transport is not yet a full physically based reflection system.

## Modern interface

The main menu, campaign, tutorial, multiplayer, options, in-game menus, and dialogs use responsive code-native layout templates. Build, Research, and Launch are right-docked managers with separate queue, quantity, cost, total, scroll, and action regions.

With ships selected, open the tactical right-click menu and choose **Ship Dossier**. The dossier shares the right dock and shows loaded ship statistics plus a concise Fleet Archive assessment.

## Fleet identity editor

The fleet palette exposes the full HSV range for primary and stripe colors, including true black. Separate switches control whether these player-only effects retain their authored colors or use a custom color:

- Engine trails and nozzle emission
- Navigation lights
- Harvesting beams
- Hyperspace transitions

The chosen effect color drives both the visible raster effect and its DXR light where that effect emits light. Enemy and allied navigation lights retain their authored identities.

## Gameplay options

These additions are opt-in and saved with the user configuration:

- **Unlimited Strike-Craft Fuel:** single-player only; normal orders and docking behavior remain available.
- **Issue Orders While Paused:** accept tactical commands while simulation time is stopped.
- **Ship Weapon Recoil:** enable physical firing impulse.
- **Cursor Scale:** 25-100% live preview.
- **Resource Multiplier:** 1x-4x deposited RU yield and collector harvest/fill rate; authored resource values are unchanged; single-player only.
- **Super Salvagers:** 2x mobility/agility/speed/braking and 3x effective health.
- **Capture / Build All:** exceptional captured production ships keep their native factory roster. Captured ships remain yours if the option is later off.

Simulation-changing convenience options are kept out of multiplayer behavior.

## Command-line examples

```
# 1440p resizable window, uncapped, no VSync
./HomeworldModern.exe /window /width 2560 /height 1440 /noVsync /fps 0

# 4K exclusive mode at 120 Hz
./HomeworldModern.exe /exclusive /width 3840 /height 2160 /refresh 120 /fps 120 /vsync

# Cross-vendor presentation without DXR lighting
./HomeworldModern.exe /noRaytracing /fxaa

# Heavier path tracing
./HomeworldModern.exe /raytracing /pathSamples 4 /pathBounces 3 /dlssQuality
```

Frequently useful switches:

| Switch                     | Meaning                                       |
|----------------------------|-----------------------------------------------|
| /window                    | Resizable window                              |
| /borderless or /fullscreen | Borderless desktop mode                       |
| /exclusive                 | Exclusive fullscreen                          |
| /width N, /height N        | Requested dimensions                          |
| /refresh N                 | Exclusive refresh; zero selects automatically |
| /fps N                     | Presentation cap; zero is uncapped            |
| /uiScale N                 | 50-200% modifier over automatic scale         |
| /raytracing, /noRaytracing | Enable/disable DXR lighting                   |
| /pathSamples N             | 1-4 samples per pixel per frame               |
| /pathBounces N             | 1-3 diffuse bounces                           |
| /fxLightStrength N         | 0-200% FX/emissive light contribution         |
| /dlaa, /fxaa, /aaOff       | AA selection                                  |

| Switch                                  | Meaning                                   |
|-----------------------------------------|-------------------------------------------|
| /dlssQuality, /dlssBalanced             | DLSS quality modes                        |
| /dlssPerformance, /dlssUltraPerformance | DLSS performance modes                    |
| /vsync, /adaptiveVsync, /noVsync        | Presentation synchronization              |
| /vanilla                                | Restore the original universe update rate |

The Video and Shift+F12 interfaces are preferred for normal use because they show allowed ranges and save the configuration.

## Files created by the game

| Location                                                           | Purpose                              |
|--------------------------------------------------------------------|--------------------------------------|
| Documents\My Games\Homeworld Modern\Homeworld.cfg                  | Saved options                        |
| Documents\My Games\Homeworld Modern\installation.ini               | Validated data location              |
| Documents\My Games\Homeworld Modern\RTXLighting\MissionNN.rtxlight | Per-user mission lighting overrides  |
| RTXExports\Global\RTX-Visual-Settings.txt                          | Portable global visual/shadow export |
| RTXExports\Missions\MissionNN.rtxlight                             | Portable mission-light export        |
| RTXExports\Maps\MissionNN.rtxmap                                   | Non-destructive map overlay export   |
| RTXExports\CloudPresets\                                           | Reusable volumetric dust presets     |
| %LOCALAPPDATA%\HomeworldModern\ShaderCache                         | Versioned presenter shader cache     |

## Diagnostics

Source-tree users can launch through:

```
./Windows/run-with-log.ps1 -Configuration Release -DataPath "D:\Games\HomeworldClassic\Data"
```

The script records startup stages, stdout/stderr, arguments, OS, GPU, display, exit code, and crash dump information beneath out/logs, then prepares a ZIP. Use -Windowed when testing whether a failure is specific to fullscreen.

For a binary-release report, provide your GPU, driver version, Windows version, mission, exact settings, and reproduction steps. Never attach retail data.

## Next reading

- [Editing Guide](#)
- [Known Limitations](#)
- [Build Guide](#)
- [Graphics Feature Status](#)



# Map, Lighting, Shadow, and Volumetric Editing

This guide covers the two live authoring workspaces included in the public beta:

- **Shift+F11:** map, object, and volumetric-dust editing
- **Shift+F12:** mission key lighting, local lights, global material tuning, and shadows

Both tools work as non-destructive overlays. They do not rewrite `Homeworld.big`, `Update.big`, HSF files, or other retail archives. Save frequently and retain the exported text files in version control.

## Safe authoring workflow

1. Keep the release in a writable folder outside `Program Files`.
2. Start the campaign mission you want to edit.
3. Open the relevant editor.
4. Work with the simulation and KAS mission script paused when changing layout.
5. Make one class of change at a time and test at normal gameplay speed.
6. Save/export from inside the editor.
7. Copy the human-readable files from `RTXExports` into your project.
8. Reload the mission and verify the overlay from disk.

The editor uses runtime IDs to refer to original mission objects. A major change to the underlying retail mission can invalidate those references; the original archive remains untouched and is always the fallback.

# Map Editor - Shift+F11

## Workspace layout

The left panel has three tabs:

- **Assets:** placeable ships, derelicts, asteroids, dust, gas, and RTX volumes.

The list is built from the table of contents of the legal BIG archives plus project procedural assets.

- **Scene:** original mission objects and objects added by the editor. Modified and deleted states are shown explicitly.

- **Volumes:** raymarched volumetric dust volumes and their material summary.

The right panel is an inspector. The top toolbar controls mission boot, pause state, viewport mode, camera preset, save/load, and transition to Lighting.

## Toolbar

| Control              | Behavior                                                               |
|----------------------|------------------------------------------------------------------------|
| Mission selector     | Click advances Mission 01-16; hold Shift while clicking to go backward |
| Load Paused          | Boots the selected mission into a clean authoring state                |
| Sim Paused / Running | Controls gameplay simulation independently                             |
| KAS Paused / Running | Controls mission scripting independently                               |
| View                 | Cycles Lit, Wireframe, Bounds, and Volumes                             |
| Camera               | Cycles Perspective, Top, Front, and Side                               |
| Save Map             | Writes the current non-destructive overlay                             |
| Load Map             | Reloads the current mission and reapplies the overlay                  |
| Lighting             | Closes the map tool and opens Shift+F12 lighting authoring             |
| Done                 | Closes the editor and restores the previous pause state                |

## Camera and selection

- Hold the right mouse button and move the mouse to aim the editor camera.
- Use the mouse wheel over the viewport to fly forward/backward through the screen-center ray.

- Click an object or a projected volume marker to select it.
- Press F to focus the camera on the selection.
- Press P in Assets to place the selected asset at cursor depth.
- Press P in Scene or Volumes to move the selection to cursor depth.

The editor intentionally avoids WASD camera movement so list/inspector input does not fight gameplay keys.

## Transforming objects

Selected world objects expose a direct 3D gizmo:

- Drag colored X/Y/Z arrowheads to translate.
- Drag blue axis handles to rotate.
- E chooses rotation mode for keyboard adjustments.

- R chooses scale mode. Scaling is limited to resources/volumes because arbitrary ship instance scaling is unsafe for gameplay systems.

- X, Y, or Z chooses the keyboard adjustment axis.
- Arrow Up/Down changes the selected inspector row.
- Arrow Left/Right changes the selected value.
- Control applies fine adjustment; Shift applies coarse translation/size steps.

Original objects can be transformed, marked for deletion, or left unchanged. Editor additions can be duplicated with `Ctrl+Shift+D`. Deletion is represented in the overlay rather than applied to the BIG archive.

## Volumetric dust

An RTX dust volume is a real D3D12 raymarched medium. Up to 128 volumes can be authored; up to 64 enabled, non-zero-density volumes are uploaded to the live renderer.

Volume fields:

| Field             | Purpose                                          | Range/notes                         |
|-------------------|--------------------------------------------------|-------------------------------------|
| Enabled           | Includes/excludes the volume without deleting it | On/Off                              |
| Shape             | Base boundary                                    | Sphere, Box, Ellipsoid              |
| Position X/Y/Z    | World-space center                               | Keyboard step 100 units             |
| Rotate X/Y/Z      | Rigid volume orientation                         | 5-degree steps; fine mode supported |
| Size X/Y/Z        | Shape extents                                    | 50-500,000 units                    |
| Density           | Medium density                                   | 0-10                                |
| Scattering        | Out-of-beam light contribution                   | 0-10                                |
| Absorption        | Light removed through the medium                 | 0-10                                |
| Anisotropy        | Forward/back scattering bias                     | -0.95 to 0.95                       |
| Coverage          | Procedural occupancy                             | 0-2                                 |
| Color R/G/B       | Linear single-scattering/albedo tint             | 0-8 per channel                     |
| Noise Scale       | World-space feature frequency                    | 0.00001-0.1                         |
| Noise Detail      | Secondary structure strength                     | 0-2                                 |
| Wake Strength     | Ship disturbance response                        | 0x off, 1x baseline, 2x maximum     |
| Mission Key Light | Receive the mission directional source           | On/Off                              |
| Local Lights      | Receive point/spot/FX lighting                   | On/Off                              |
| Volume Shadow     | Cast volumetric shadowing                        | On/Off                              |
| Shape Seed        | Deterministic silhouette seed                    | G randomizes                        |
| Shape Variation   | Boundary deformation                             | 0-2                                 |

Tune in this order: size and transform, coverage/noise, density, scattering and absorption, color, lighting flags, then wake response. Very high density plus large overlapping volumes can obscure the scene and sharply increase raymarch cost.

## Cloud preset library

The Volumes tab exposes a reusable preset dropdown:

- **Save New:** stores the selected volume's complete material, shape, seed, transform defaults, and lighting flags as a new preset.
- **Overwrite:** replaces the selected preset from the current volume.
- **Load:** loads the preset into the selected volume or creates a new volume.
- **Copy / Paste:** duplicates full volumes between locations; Ctrl+C and Ctrl+V provide the same operation.

Presets and their index live beneath:

```
RTXExports\CloudPresets\
```

## Map overlay format and path

Save Map writes:

```
RTXExports\Maps\MissionNN.rtxmap
```

The RTXMAP 8 text format contains:

- MISSION identity
- ADD records for placed ships/resources
- TRANSFORM records for moved/rotated/scaled objects
- DELETE records for hidden original objects
- VOLUME DUST records for raymarched media

The packaged assets\Missions\MissionNN.rtxmap is used when no writable export exists. Writable exports take precedence, which makes iteration immediate.

Do not hand-edit runtime IDs unless you understand the mission object order. Hand editing material values is possible, but the live inspector is safer and enforces limits.

# Lighting Editor - Shift+F12

The lighting editor does not pause gameplay. Open it after the mission HDR sky has loaded. Use Tab or click tabs to move between **Mission Key**, **Custom Lights**, **Global Tuning**, and **Shadows**.

Rows can be adjusted by mouse wheel, +/- buttons, or Arrow Left/Right. Arrow Up/Down selects a row.

## Mission Key page

| Field                  | Meaning                                                   |
|------------------------|-----------------------------------------------------------|
| Sun Azimuth            | Rotates the key around world Y; 0 degrees points along -Z |
| Sun Elevation          | Raises/lowers the key, limited to -89 to +89 degrees      |
| Sun Intensity          | Per-mission multiplier, 0-4 in 0.05 steps                 |
| Sun Red/Green/Blue     | Multipliers over detected HDR source color, 0-4           |
| Ambient Strength       | HDR-derived fill strength, 0-4                            |
| Ambient Color Mode     | Auto follows HDR sky mean; Manual uses RGB below          |
| Ambient Red/Green/Blue | Manual linear ambient tint, 0-4                           |

P aims the mission key using the cursor ray. I temporarily solos the key by muting sky ambient, environment, authored lights, and FX. R restores the HDR metadata baseline and reimports retail map lights. These changes are live and remain unsaved until S or **Save Mission**.

## Custom Lights page

When the editor opens, retail HSF local lights are adopted into the editable list. System key and ambient entries remain identifiable; editable records may be Point, Spot, or Ambient.

| Field            | Meaning                                                         |
|------------------|-----------------------------------------------------------------|
| Select Light     | Cycles key, sky ambient, imported HSF lights, and added lights  |
| Light Type       | Point, Spot, or Ambient for editable records                    |
| Enabled          | Temporarily bypasses a light without deleting it                |
| Intensity        | Per-light energy, also affected by Mission-Authored global gain |
| Red/Green/Blue   | Linear light color                                              |
| Range / Radius   | Point/spot falloff distance in world units                      |
| Spot Cone        | Inner spot half-angle                                           |
| Spot Edge        | Feather outside the inner cone                                  |
| Y Rotation / Yaw | Rotates a spot around world Y                                   |
| Position X/Y/Z   | Fine world-space placement                                      |

Shortcuts:

- N adds a light.
- DeLeTe removes the selected editable light.
- P places the selected point/spot at cursor depth.
- A aims the selected spot at the cursor.
- S saves the mission override and portable export.
- L reloads the saved override, then adopts retail map lights when needed.

Use the on-screen world marker to verify position and direction. Prefer a small number of deliberate lights; each shadow-casting source adds visibility work.

## Global Tuning page

| Control                  | Range   | What it changes                                            |
|--------------------------|---------|------------------------------------------------------------|
| Primary Sun / Shadow     | 0-200%  | Gain after mission-local key settings                      |
| Sky Ambient / Color Wash | 0-200%  | Mission ambient fill                                       |
| Environment Reflections  | 0-200%  | Sky sampling and neutral path floor                        |
| Mission-Authored Lights  | 0-200%  | HSF and live authored lights                               |
| FX / Emissive Lights     | 0-200%  | Weapons, engines, beams, explosions, glow maps, nav lights |
| Surface Reflectivity     | 0-200%  | Stable primary-surface highlights                          |
| Surface Roughness        | 0-100%  | Highlight spread/sharpness                                 |
| Generated Normals        | 0-400%  | LIF-derived relief                                         |
| Path-Light Exposure      | 50-200% | Final path-light transfer only                             |
| Color Banding Filter     | 0-200%  | Stable scene dithering before 8-bit presentation           |

R restores the tuned defaults: 100% light groups, 125% reflectivity, 38% roughness, 100% generated normals, 100% exposure, and 100% dithering. E saves options and exports all portable global visual/shader settings.

## Shadows page

| Control                 | Range            | Baseline | Guidance                                        |
|-------------------------|------------------|----------|-------------------------------------------------|
| Shadow Strength         | 0-200%           | 100%     | Global opacity of ray-traced visibility         |
| Sun Angular Radius      | 0-5 degrees      | 0.35     | Larger values broaden distant penumbrae         |
| Sun Shadow Samples      | 1-16             | 6        | Higher steadiness at higher cost                |
| Local Shadow Samples    | 1-8              | 2        | Point/spot/FX visibility rays                   |
| Local Light Radius      | 0-10 degrees     | 0.20     | Area spread for local sources                   |
| Receiver Bias           | 0.01-2.00        | 0.05     | Outgoing-ray offset; fights acne                |
| Normal Bias             | 0-4.00           | 0.08     | Surface-normal offset; fights self-intersection |
| Contact Shadow Strength | 0-200%           | 100%     | Near-field reinforcement beneath soft shadows   |
| Contact Distance        | 0-10,000         | 1,200    | Range of contact reinforcement                  |
| Max Shadow Distance     | 10,000-1,000,000 | 500,000  | Visibility-ray reach                            |

Shadow tuning order:

1. Set sun and local sample counts for your performance target.
2. Set sun/local angular radius for the desired softness.
3. Lower receiver and normal bias until acne appears, then increase slightly.
4. Tune contact strength/distance to restore grounding lost to soft shadows.
5. Shorten maximum distance only when the mission scale permits it.

Too little bias creates self-shadow acne; too much disconnects shadows from their casters (peter-panning). High sample counts multiply path-tracing cost. R resets only the Shadows page when it is active. E exports shadows with the global visual state.

## Lighting save and export paths

S or **Save Mission** writes both:

```
Documents\My Games\Homeworld Modern\RTXLighting\MissionNN.rtxlight  
RTXExports\Missions\MissionNN.rtxlight
```

The first is the live per-user override. The second is the portable file to commit or share. The format begins with RTXMISSIONLIGHTING 3 and records key direction/energy, ambient state, whether authored lights replace retail map lights, and every editable local light.

E or **Export Visuals** writes:

```
RTXExports\Global\RTX-Visual-Settings.txt
```

This contains display timing, path settings, image quality, post effects, global tuning, and shadow settings. Resolution, refresh rate, AA/DLSS mode, and frame generation are deliberately excluded so an author's export does not force hardware-specific choices on another user.

# Shortcut reference

## Map Editor

| Key                | Action                                                 |
|--------------------|--------------------------------------------------------|
| Shift+F11 or Esc   | Close editor                                           |
| Right mouse + move | Aim camera                                             |
| Mouse wheel        | Fly forward/back                                       |
| P                  | Place selected asset or move selection to cursor depth |
| F                  | Focus selection                                        |
| E                  | Rotation mode                                          |
| R                  | Scale mode for resources/volumes                       |
| X, Y, Z            | Select adjustment axis                                 |
| Arrow Up/Down      | Select inspector row                                   |
| Arrow Left/Right   | Adjust value                                           |
| Ctrl               | Fine adjustment modifier                               |
| Shift              | Coarse translation/size modifier                       |
| Delete             | Delete/hide selection through overlay                  |
| Ctrl+Shift+D       | Duplicate selection                                    |
| G                  | Randomize selected dust silhouette                     |
| Ctrl+C, Ctrl+V     | Copy/paste full dust volume                            |
| Ctrl+S             | Save map overlay                                       |
| L                  | Reload mission and overlay                             |

## Lighting Editor

| Key              | Action                                        |
|------------------|-----------------------------------------------|
| Shift+F12 or Esc | Close editor and save global options          |
| Tab              | Cycle Mission, Lights, Tuning, Shadows        |
| Arrow Up/Down    | Select row                                    |
| Arrow Left/Right | Adjust live value                             |
| P                | Place/aim key or place selected local light   |
| A                | Aim selected spot light                       |
| N                | Add local light                               |
| Delete           | Delete selected editable local light          |
| I                | Solo mission key                              |
| R                | Reset active authoring/tuning page            |
| S                | Save and export mission lighting              |
| L                | Reload saved mission lighting                 |
| E                | Save and export global visual/shadow settings |



# Publishing authored work

Share only .rtxmap, .rtxlight, cloud-preset, and visual-setting text files plus art you have the right to redistribute. Never include Homeworld.big, Update.big, voice/music archives, movies, or extracted retail assets.

For a packaged project preset, keep this layout:

```
Missions\MissionNN.rtxmap
RTXLighting\MissionNN.rtxlight
CloudPresets\index.txt
CloudPresets\*.rtxdust
```

Document the required mission/data edition, gameplay-affecting object changes, and the graphics cost of high-volume/high-light scenes.

# Known Limitations and Beta Expectations

Homeworld RTX 0.91.3 Beta is a working public test build, not a finished remaster. The following boundaries are intentional and should be understood before use.

## Platform

- Windows 10/11 x64 only.
- The active renderer depends on Direct3D 12, DXGI, DXR, and Windows tooling.
- There are no supported Linux, macOS, Steam Deck, or ARM64 binaries.
- DXR lighting requires capable hardware; NVIDIA DLAA/DLSS requires a compatible

NVIDIA RTX GPU and driver.

## Required retail data

- The release contains no retail BIG archives, speech, music, or movies.
- A legal Homeworld Classic installation is required.
- Alternate language and patched data layouts are validated structurally, but the full matrix of editions has not been campaign-tested.
- Do not combine Remastered Collection Classic data with a separate original Update.big.

## Renderer

- Native D3D12 fixed-function compatibility rasterization is experimental and is not parity-complete for every rare legacy state.
- HDR10 and sRGB output are not implemented; output is SDR.
- Ray Reconstruction is not implemented.
- Frame Generation is disabled. The retired generic interpolation experiment is not presented as NVIDIA DLSS-G.
- NVIDIA Reflex markers/latency integration are not implemented.
- Full specular reflection transport and authored metalness/roughness materials are not implemented. Classic materials use deterministic approximations.
- Animated/morphed BLAS updates and uncommon transparency/reactive cases may still need work.
- Multiple importance sampling and a production neural denoiser are not present.
- Very high path sample, bounce, shadow-ray, light-count, and volumetric settings can be expensive even on recent GPUs.
- The shader cache is persisted, but driver-specific D3D12 pipeline-library blobs are deliberately not reused because cross-run driver behavior was unsafe.

## Visual behavior

- The renderer forces LOD0 and removes N-LIPS distance enlargement. Dense fleets may therefore cost more GPU time than the original game.
- Path-traced lighting converges progressively. Fine noise can be visible after

cuts, fast movement, visibility changes, or at low sample counts.

- Fast motion intentionally reduces temporal history to avoid ghosting.
- Generated normals are inferred from legacy LIF luminance and cannot reproduce

hand-authored modern normal maps.

- Post effects are artistic additions and default conservatively; they may not match every mission palette.

- UI plate crops and ultrawide safe-canvas composition still require broad localization and DPI review.

## Gameplay and multiplayer

- The project aims to preserve the fixed simulation contract, but this beta has not been exhaustively checksum-tested across every campaign and multiplayer scenario.

- Unlimited fuel, resource multiplier, Super Salvagers, and Capture / Build All are single-player convenience features and are excluded or constrained for multiplayer safety.

- Legacy online service code is not a supported matchmaking service. Treat LAN functionality as experimental and use identical builds/settings where required.

- Save compatibility follows the HomeworldSDL baseline; retain backups before relying on a beta for a long campaign.

## Editors

- Shift+F11 and Shift+F12 are development tools and assume familiarity with mission scale, object types, and lighting tradeoffs.

- Map overlays depend on runtime object IDs. Changes to the underlying mission can invalidate transformations/deletions.

- Only 64 enabled non-zero-density volumetric dust volumes are uploaded to the live raymarcher, though up to 128 can be authored.

- Extremely large/overlapping volumes, high density, high shadow counts, and many local lights can severely reduce performance.

- The editors do not provide undo history. Save versions of exported text files.

- Mission and KAS pause states are independent; changing a live mission without pausing scripts can produce moving targets or script-side changes.

- Export directories must be writable.

## Content and support

- This is a free, non-commercial community project.

- It is not affiliated with or supported by Gearbox, Relic, Sierra, NVIDIA, Steam, or the original developers/publishers.

- Do not report project bugs to the commercial game-support teams.

- Do not upload copyrighted retail data in bug reports.

- No warranty is provided. The Relic source-code agreement in `LICENSE.txt` governs the original source baseline and includes restrictive terms.

## Useful report checklist

Before opening an issue, record:

1. Windows edition/build.
2. GPU model and driver version.
3. Homeworld data edition and language.
4. Mission/mode and save provenance.
5. Display mode, resolution, refresh, VSync, and frame cap.
6. Path-tracing, sample, bounce, AA/DLSS, shadow, and post-effect settings.
7. Exact steps and whether the issue reproduces after renderer reset/restart.
8. Diagnostic logs or dump, without retail assets.

# Third-Party and Content Notices

Homeworld RTX combines an original Homeworld/HomeworldSDL source baseline with new project code and dynamically distributed runtime libraries. Every component retains its own license; this notice is a guide, not a replacement for those terms.

## Homeworld source baseline

The original Homeworld source and modified software are governed by the Relic Entertainment source-code agreement reproduced in `LICENSE.txt`. Its grant is limited to non-commercial use/distribution and includes additional restrictions. The project and binaries are provided free of charge.

Homeworld, Relic, Sierra, Gearbox, and related names and marks belong to their respective owners. No affiliation or endorsement is claimed.

Retail data is not distributed. Users must supply their own legal installation. Do not redistribute `Homeworld.big`, `Update.big`, `HW_Comp.vce`, `HW_Music.wxd`, or retail movies through this project.

## SDL2 and SDL2\_net

SDL2 and SDL2\_net are used for platform, input, audio, and optional networking services. The Windows release dynamically loads `SDL2.dll`. Source and current license information are available from:

- <https://www.libsdl.org/>
- <https://github.com/libsdl-org/SDL>
- [https://github.com/libsdl-org/SDL\\_net](https://github.com/libsdl-org/SDL_net)

The exact dependency versions are resolved by the repository's `vcpkg.json` and `vcpkg` baseline at build time.

## FFmpeg

FFmpeg libraries provide Bink movie demuxing/decoding, timing, and video pixel conversion. The release dynamically loads `avcodec`, `avformat`, `avutil`, and `swscale` DLLs. FFmpeg is generally available under LGPL 2.1 or later, but the effective license depends on build options and external libraries.

- Project: <https://ffmpeg.org/>
- Source: <https://github.com/FFmpeg/FFmpeg>
- License overview: <https://github.com/FFmpeg/FFmpeg/blob/master/LICENSE.md>

The distributed DLLs are generated by the `vcpkg` manifest build. Corresponding source recipes, build configuration, and upstream source references are available through the pinned `vcpkg` dependency process used by this repository. No `--enable-nonfree` configuration is requested by this project.

## NVIDIA Streamline, DLSS, and NGX runtimes

NVIDIA Streamline 2.12.0 is integrated for DLAA and DLSS reconstruction. The bootstrap downloads NVIDIA's signed SDK archive from its official GitHub release, validates a pinned SHA-256 digest, and copies the required production runtimes. NVIDIA binaries are not committed to the source repository; selected redistributable DLLs are included with the Windows binary release.

- Streamline: <https://github.com/NVIDIA-RTX/Streamline>
- Programming guide: <https://github.com/NVIDIA-RTX/Streamline/blob/main/docs/ProgrammingGuide.md>
- NVIDIA RTX SDK license: included as `licenses/NVIDIA-Streamline-license.txt`
- NVIDIA NGX/DLSS license: included as `licenses/NVIDIA-DLSS-license.txt`

NVIDIA, RTX, DLSS, and related names are trademarks of NVIDIA Corporation.

## stb headers

src/ThirdParty/stb contains single-header libraries from the stb project. Their dual MIT/public-domain notices are embedded in the source headers.

- <https://github.com/nothings/stb>

## DirectX Shader Compiler and Windows SDK

DXC is a build-time dependency used to compile the DXR HLSL library. Windows, Direct3D, DirectX, DXGI, and related SDK components belong to Microsoft and are used under their applicable SDK/runtime terms.

- <https://github.com/microsoft/DirectXShaderCompiler>

## Project artwork and generated assets

Provenance notes are located beside relevant assets:

- assets/UI/ART\_PROVENANCE.md
- assets/Cursors/CURSOR\_PROVENANCE.md
- assets/Missions/SKY\_PROVENANCE.md

Some interface/cursor/mission imagery is restored, transformed, or generated from lawfully supplied Homeworld material. Distribution does not grant rights to extract or reuse that material outside this non-commercial Homeworld project.

## No endorsement

References to companies, products, and APIs identify compatibility only. This project is community-developed, unsupported by those companies, and provided without warranty.